CCNA 200-301
Master Question Bank

Curated High-Yield Q&A for Cisco Ideathon Prep

This document compiles the most critical and frequently tested concepts across all CCNA domains: Network Fundamentals, Network Access, IP Connectivity, IP Services, Security, and Automation. Optimized for rapid revision and deep technical understanding.

Domain 1: Network Fundamentals

1. What is the primary function of a router at the Network Layer (Layer 3)?

A router's primary function is to forward data packets between different computer networks. It uses IP addresses to determine the best path to route packets to their final destination, breaking up broadcast domains.

2. Differentiate between TCP and UDP.

TCP (Transmission Control Protocol) is connection-oriented, ensuring reliable delivery through acknowledgments, sequencing, and flow control. UDP (User Datagram Protocol) is connectionless, offering "best-effort" delivery without error recovery, making it faster and ideal for voice/video streaming.

3. What are the private IP address ranges for IPv4 as defined by RFC 1918?

Class A: 10.0.0.0 to 10.255.255.255 (Prefix 10.0.0.0/8). Class B: 172.16.0.0 to 172.31.255.255 (Prefix 172.16.0.0/12). Class C: 192.168.0.0 to 192.168.255.255 (Prefix 192.168.0.0/16).

4. Explain the concept of a Default Gateway.

A default gateway is the node on a network that serves as the forwarding host to other networks when no other route specification matches the destination IP address of a packet.

5. How does IPv6 address the IPv4 exhaustion problem?

IPv6 uses a 128-bit address space, compared to IPv4's 32-bit address space, providing approximately 340 undecillion addresses, virtually eliminating the risk of address exhaustion.

6. What is the role of ARP in a network?

The Address Resolution Protocol (ARP) resolves a known IPv4 address to an unknown MAC (Media Access Control) address on a local area network (LAN), allowing packets to be encapsulated into frames for physical delivery.

7. Describe the function of the Transport Layer in the OSI model.

The Transport Layer (Layer 4) provides end-to-end communication services for applications, including segmentation, multiplexing (using port numbers), connection establishment, and error recovery (in the case of TCP).

8. What is the difference between a broadcast domain and a collision domain?

A collision domain is a network segment where data packets can collide if sent simultaneously (separated by switches). A broadcast domain is a logical division of a network where all nodes can reach each other via broadcast at the data link layer (separated by routers or VLANs).

9. What are the three types of IPv6 addresses?

Unicast (one-to-one), Multicast (one-to-many), and Anycast (one-to-nearest). Note that IPv6 does not use broadcast addresses.

10. Explain the role of a hypervisor in virtualization.

A hypervisor (or Virtual Machine Monitor) is software that creates and runs virtual machines. It abstracts underlying physical hardware resources (CPU, RAM, storage) and allocates them to independent virtual machines.

Domain 2: Network Access

11. What is the purpose of a VLAN?

A Virtual Local Area Network (VLAN) logically segments a physical network into multiple broadcast domains. This improves security, simplifies management, and enhances network performance by reducing broadcast traffic.

12. Explain the function of IEEE 802.1Q.

802.1Q is the IEEE standard for VLAN tagging. It inserts a 4-byte tag into the Ethernet frame header to identify the VLAN to which the frame belongs when traversing trunk links.

13. What problem does the Spanning Tree Protocol (STP) solve?

STP (IEEE 802.1D) prevents Layer 2 loops in redundant network topologies. It achieves this by electing a Root Bridge and blocking redundant paths, ensuring there is only one active logical path between any two switches.

14. What are the different port states in Rapid Spanning Tree Protocol (RSTP)?

RSTP (802.1w) simplifies the port states to three: Discarding, Learning, and Forwarding. (It merges the traditional STP Blocking and Listening states into Discarding).

15. How does PortFast benefit a switch port?

PortFast is an STP feature applied to access ports connected to end devices. It allows the port to bypass the Listening and Learning states and transition immediately to the Forwarding state, speeding up network access for endpoints.

16. What is EtherChannel and what protocol is Cisco proprietary for it?

EtherChannel bundles multiple physical Ethernet links into a single logical link for increased bandwidth and redundancy. PAgP (Port Aggregation Protocol) is Cisco proprietary, while LACP (Link Aggregation Control Protocol) is the IEEE 802.3ad standard.

17. Describe the function of a Wireless LAN Controller (WLC).

A WLC manages lightweight access points (LAPs) in large-scale wireless networks. It centralizes configuration, security policies, RF management, and client roaming, using the CAPWAP protocol to communicate with APs.

18. What is the difference between an Access Port and a Trunk Port?

An access port belongs to and carries traffic for only one VLAN (untagged). A trunk port acts as a point-to-point link between switches or a switch and a router, carrying traffic for multiple VLANs using 802.1Q tagging.

19. What is a Native VLAN?

In an 802.1Q trunk, the native VLAN is the only VLAN whose frames are sent untagged. Both ends of a trunk link must have the same native VLAN configured to avoid traffic leaking or STP issues.

20. What role does CSMA/CD play in Ethernet?

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) was used in half-duplex Ethernet to detect and manage data collisions. In modern full-duplex switched networks, CSMA/CD is largely obsolete as collisions do not occur.

Domain 3: IP Connectivity

21. Define Administrative Distance (AD).

Administrative Distance is a metric used by Cisco routers to select the best routing protocol when there are multiple paths to the same destination from different routing protocols. A lower AD indicates higher reliability (e.g., Connected=0, Static=1, OSPF=110).

22. What is a floating static route?

A floating static route is a backup route configured with a higher administrative distance than the primary route (such as an OSPF or EIGRP route). It only enters the routing table if the primary route fails.

23. Explain how OSPF calculates its metric.

OSPF uses "Cost" as its metric, which is inversely proportional to the bandwidth of the link. The default formula is Reference Bandwidth (100 Mbps) / Interface Bandwidth.

24. What is the purpose of the OSPF Router ID?

The Router ID uniquely identifies an OSPF router within an autonomous system. It is determined by the highest active IP address on a loopback interface, or if none exists, the highest IP address on an active physical interface (unless manually configured).

25. Distinguish between an interior gateway protocol (IGP) and an exterior gateway protocol (EGP).

IGPs (like OSPF, EIGRP, IS-IS) route traffic within a single Autonomous System (AS). EGPs (like BGP) route traffic between different Autonomous Systems across the internet.

26. What does a routing table longest prefix match mean?

When a router forwards a packet, it compares the destination IP to the routing table. If multiple routes match, the router chooses the route with the longest subnet mask (longest prefix), as it is the most specific route to the destination.

27. Describe an IPv6 Link-Local address.

An IPv6 link-local address (starting with FE80::/10) is automatically generated on every IPv6-enabled interface. It is used for communications strictly within the local subnet and is not routable.

28. What is the First Hop Redundancy Protocol (FHRP)?

FHRP provides high availability for default gateways by allowing multiple routers to share a single virtual IP address and MAC address. Examples include HSRP (Cisco), VRRP (Standard), and GLBP.

29. How does OSPF establish neighbor adjacencies?

OSPF routers send Hello packets on OSPF-enabled interfaces. If two routers agree on parameters (Area ID, Hello/Dead intervals, Authentication, Stub area flag), they form an adjacency to exchange Link-State Advertisements (LSAs).

30. What is a Default Route?

A default route (0.0.0.0/0 in IPv4) is a "catch-all" route used when the destination IP address does not match any other specific entry in the routing table.

Domain 4: IP Services

31. What is the function of NAT?

Network Address Translation (NAT) translates private, non-routable IP addresses (RFC 1918) into a public, routable IP address for internet access, conserving the limited pool of public IPv4 addresses.

32. Differentiate between Static NAT and PAT.

Static NAT maps a single private IP to a single public IP (1:1). Port Address Translation (PAT), or NAT Overload, maps multiple private IPs to a single public IP by tracking unique source port numbers (Many:1).

33. How does DHCP function using the DORA process?

Dynamic Host Configuration Protocol (DHCP) assigns IPs automatically. DORA stands for Discover (client broadcasts request), Offer (server offers IP), Request (client accepts offer), Acknowledge (server confirms lease).

34. What is the role of DNS in a network?

The Domain Name System (DNS) resolves human-readable hostnames (e.g., www.cisco.com) into IP addresses that networking equipment uses to route traffic.

35. Why is NTP critical for network operations?

Network Time Protocol (NTP) synchronizes the clocks of network devices. Accurate time is critical for correlating syslogs, verifying digital certificates, and ensuring time-dependent protocols function correctly.

36. What is Syslog and what are its severity levels?

Syslog is a standard for message logging. Cisco severity levels range from 0 (Emergency) to 7 (Debugging). Lower numbers indicate more critical events.

37. Explain the function of SNMP.

Simple Network Management Protocol (SNMP) is used to monitor, configure, and manage network devices. SNMPv3 is the current standard, offering encryption, message integrity, and authentication.

38. What is a DHCP Relay Agent (IP Helper)?

A DHCP Relay Agent forwards DHCP broadcast messages from clients on a local subnet to a DHCP server on a different subnet, transforming the broadcasts into unicast messages.

39. What is QoS and why is it used?

Quality of Service (QoS) manages network resources by prioritizing certain types of traffic (like voice or video) over others (like file transfers) to prevent latency, jitter, and packet loss on congested links.

40. Define SSH and its advantage over Telnet.

Secure Shell (SSH) is a cryptographic network protocol for operating network services securely over an unsecured network. Unlike Telnet, which sends data in plaintext, SSH encrypts all traffic, including passwords.

Domain 5 & 6: Security, Automation, & Programmability

41. What is an Access Control List (ACL)?

An ACL is a set of rules defined on a router or firewall that permits or denies network traffic based on source/destination IPs, protocols, and port numbers, effectively filtering traffic.

42. Differentiate between Standard and Extended ACLs.

Standard ACLs (1-99) filter traffic based solely on the source IPv4 address. Extended ACLs (100-199) filter based on source/destination IPs, protocols (TCP/UDP/ICMP), and port numbers.

43. What is Port Security on a switch?

Port Security restricts a switch port to only accept traffic from specific MAC addresses. If an unauthorized MAC address connects, the switch can take action, such as shutting down the port (violation shutdown).

44. Explain the AAA framework.

Authentication (who you are), Authorization (what you are allowed to do), and Accounting (tracking what you did). Protocols like RADIUS and TACACS+ implement AAA in Cisco environments.

45. What is a Site-to-Site VPN?

A Site-to-Site Virtual Private Network connects entire networks (e.g., a branch office to a headquarters) over the internet using IPsec to encrypt and secure all data passing between the sites.

46. What is Software-Defined Networking (SDN)?

SDN separates the network control plane (decision making) from the data plane (forwarding). A centralized SDN controller dynamically manages network behavior via APIs, improving agility and automation.

47. What are Northbound and Southbound APIs in SDN?

Northbound APIs communicate between the SDN Controller and upper-level applications/orchestration tools. Southbound APIs (like OpenFlow or NETCONF) communicate between the controller and the physical networking devices.

48. Compare JSON and XML data formats.

Both are data serialization formats. JSON (JavaScript Object Notation) uses key-value pairs and arrays, is lightweight, and easily readable. XML uses hierarchical tags and is generally more verbose. JSON is preferred in modern REST APIs.

49. What is a REST API?

Representational State Transfer (REST) is an architectural style for APIs that uses standard HTTP methods (GET, POST, PUT, DELETE) to interact with resources, usually returning data in JSON format.

50. Briefly explain Puppet, Chef, and Ansible.

These are configuration management tools. Puppet uses a declarative language with an agent on devices. Chef uses imperative Ruby "recipes." Ansible uses YAML playbooks, operates agentless via SSH, and is highly popular for network automation.